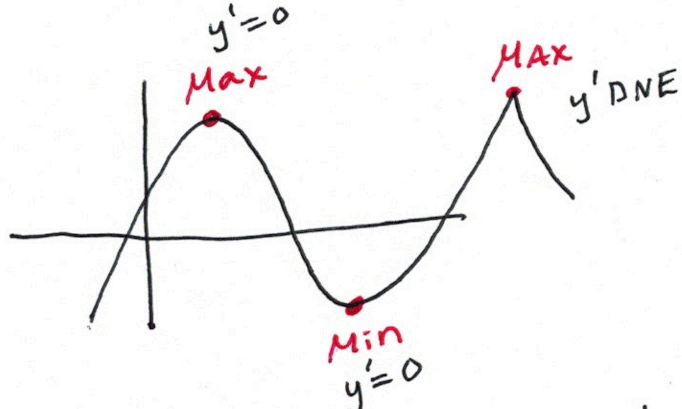
Section 4.1 Minimum & Maximums



Find x's so that $f'(x) = 0$ or $f'(x)$ DNE

These x's are critical values.

Ex① $f(x) = x^3 - x$

$$f'(x) = 3x^2 - 1$$

$$0 = 3x^2 - 1$$

$$1 = 3x^2$$

$$\frac{1}{3} = x^2$$

$$\pm\sqrt{\frac{1}{3}} = x$$

$$x = \pm\frac{1}{\sqrt{3}}$$

MAX $\left(-\frac{1}{\sqrt{3}}, 0.385\right)$

Min $\left(\frac{1}{\sqrt{3}}, -0.385\right)$

FACT If $f(x)$ has a min or max, they will occur at critical values.

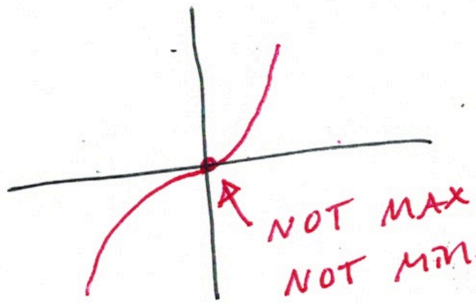
Ex ② $f(x) = x^3$

$f'(x) = 3x^2$

$0 = 3x^2$

$0 = x^2$

$0 = x$



EX ③ $y = \dfrac{2x+1}{x^2+1}$

$$y' = \frac{2(x^2+1) - (2x+1)(2x)}{(x^2+1)^2}$$

$$y' = \frac{2x^2+2 - 4x^2 - 2x}{(x^2+1)^2}$$

$$y' = \frac{-2x^2 - 2x + 2 = 0}{(x^2+1)^2 = 0}$$

<u>y' DNE</u>

$(x^2+1)^2 = 0$

$x^2+1 = 0$

$\sqrt{x^2} = \sqrt{-1}$

No sol.

<u>$y' = 0$</u>

$-2x^2 - 2x + 2 = 0$

$$x = \frac{-(-2) \pm \sqrt{\overset{4+16}{(-2)^2 - 4(-2)(2)}}}{2(-2)}$$

$$x = \frac{2 \pm \sqrt{20}}{-4}$$

$x = -1.618$

$x = 0.618$

<u>Min</u> $(-1.618, -0.618)$

<u>Max</u> $(0.618, 1.618)$

EX (4) $f(x) = \sin x + \cos x$ on $[0, 2\pi]$

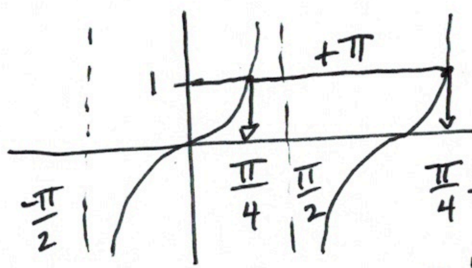
$$f'(x) = \cos x - \sin x$$

$$0 = \cos x - \sin x$$

$$\frac{\sin x}{\cos x} = \frac{\cos x}{\cos x}$$

$$\tan x = 1$$

$$x = \frac{\pi}{4}, \quad x = \frac{5\pi}{4}$$



$$\frac{\pi}{4} + \frac{\pi \cdot 4}{4} = \frac{5\pi}{4}$$

Max $(0.785, 1.414)$

Min $(3.927, -1.414)$

Ex (5) $y = e^{-x} - e^{-2x}$ on $[0,1]$

$$y' = e^{-x}(-1) - e^{-2x}(-2)$$

$$y' = \frac{-1}{e^{x}} + \frac{2}{e^{2x}}$$

$$e^{2x} \cdot 0 = \left(-\frac{1}{e^{x}} + \frac{2}{e^{2x}}\right) e^{2x}$$

$$0 = -e^{x} + 2$$

$$\ln e^{x} = \ln 2$$

$$x = \ln 2$$